A to Z of data recovery

Hanu Prateek
readersletters@thinkdigit.com

In today's world, almost every bit of information is becoming electronic. While on one hand, this helps make life easier, simpler and faster; on the other hand, if proper care isn't taken you'll have hell to pay if your data is lost. Losing important files and data, be it business documents, important photos of friends and family or even personal media like movies and songs can be very stressful, to say the least. In most cases, data is lost forever. There are still ways by which you can retrieve data or replace it with a backup taken in advance.

Prevention is better than cure

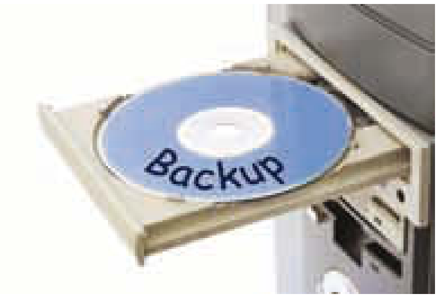
It might seem intuitive but first and foremost, the best way to recover data is to already have your important data backed up on another storage device. Care needs to be taken about how often you back up because this entails the requirement of additional storage space and time for frequent backups. You can take a backup

Recovering your lost data

on the same computer (same drive or a different partition), a network server or on the internet (an external server or the cloud). In case of any data loss, recovery then becomes a simple matter of copying back the files. In the worst case scenario of your computer crashing, you'll just have to reinstall your applications and replace user data which has been backed up.

Fixing your hardware

In case of an unavailability of such a backup and hardware problems with your disks, there are specialized organisations and service providers that can retrieve data from damaged computers and disks. This usually involves them having to move the drive to a working computer, or opening it and replacing its



Back up your data

internal parts such as read/write heads, actuator arms and chips. Sometimes, the platters are removed and placed into another drive. Of course, these types of recovery cost you a bomb on a per MB basis! Fear not though, Digit's looking out for you and we've included a DIY in the following pages detailing how you could recover data yourself.

Retrieving deleted files

Recovering a file that has been deleted recently is a relatively simple task. There are dozens of undelete utilities available that help you do just that. But two software we'll try and explore in this article are Photorec for photos and Recuva

External services can fix your hardware

from Piriform, for just about anything. Recuva is free and very simple to use offering sector-based recovery. It's also available in a lite portable version.

Sector-based recovery

When you face a hard drive crash, or your PC keeps producing error messages and problems like not being able to boot into your operating system, you need more than just a simple undelete utility. First check if the crashed drive is visible in the Windows Disk Management console, and is still spinning and working (not making any weird sounds). In this case you may still be able to retrieve your data with a sector-based utility. Unfortunately, most sector-based recovery utilities aren't available for free. When using a sector-based recovery utility keep in mind that it shouldn't be installed on the hard drive or partition from where you're trying to recover the data. You might accidentally end up overwriting that particular data. If you have a recent backup of your data and need to recover only one small important file, you can use Easeus Data Recovery Wizard. Its demo version can recover one file for you irrespective of its size. It's probably one of the easiest recovery programs to use. R-Studio is another utility which works with almost every file system out there, including Linux and Mac. The demo comes with its own boot CD in case you need to recover from a machine that can't boot to its OS. "Active File Recovery" is another software with a very user-friendly and simple interface making it easier to see what files have been recovered compared to R-Studio. In most cases, after restoring a partition you should be able to boot to the OS again failing which you'll at least be able to copy files to recover your data. For doing this, you'll probably have to boot your computer using a live CD. Any versions of Linux that will run from a CD, such as Slax or Ubuntu are adequate and they're free.